

A Computational Approach to Identifying Evolutionarily Stable Strategies in Mammalian Search Behaviour

Lynndsay A. Terpsma^{1,2}, Ricardo Martinez-Garcia^{3,4,5}, Christen H. Fleming⁶, William F. Fagan⁷, Justin M. Calabrese^{3,7,8}, and Michael J. Noonan^{1,2,9}

¹Department of Biology, University of British Columbia Okanagan, Kelowna, British Columbia, Canada

²Institute for Biodiversity, Resilience, and Ecosystem Services, University of British Columbia Okanagan, Kelowna, British Columbia, Canada

³Center for Advanced Systems Understanding (CASUS) – Helmholtz-Zentrum Dresden-Rossendorf (HZDR), Untermarkt 20, Görlitz 02826, Germany

⁴CTP South American Institute for Fundamental Research & Instituto de Física Teórica, Universidade Estadual Paulista - UNESP, São Paulo SP, Brazil

⁵Department of Ecology, Institute of Biosciences, University of São Paulo, São Paulo, R. do Matão, 321 - Butantã, 05508-090, São Paulo, Brazil

⁶Department of Biology, University of Central Florida, Orlando, FL, USA

⁷Department of Biology, University of Maryland, Maryland, USA

⁸Department of Ecological Modelling, Helmholtz Centre for Environmental Research — UFZ, Leipzig, Germany

⁹Department of Computer Science, Math, Physics, and Statistics, University of British Columbia Okanagan, Kelowna, British Columbia, Canada

Abstract

Understanding how organisms navigate their environment under landscape changes is increasingly important under rapid human-induced changes, yet predictive models often fail to incorporate ecological constraints which govern animal movement. This study aims to investigate how mammalian movement strategies adapt to variation in resource distribution using a computational approach. We developed a stochastic system, grounded in allometric scaling relationships, where individual movement behaviour emerges from interaction with variable resource landscapes and energetic costs, evolving over evolutionary time. Our results indicate that search strategies in a population stabilise over time, with emergent optima dependent on landscape composition. A fundamental trade-off between search efficiency and movement speed, emerges as a result of energetic constraints. Higher resource availability reduced ballistic length scale (l_v) (Gamma GLM: $\beta = -1.35 \times 10^{-4}$, $p = 0.00448$) and increased movement speed non-linearly (Nonlinear least squares (Michaelis-Menten): $a = 5.22$, $p < 0.001$, $b = 1079$, $p < 0.001$), suggesting individuals rely less on slow long-distance movement

when resources are energetically rich. In contrast, when resources are highly clustered, we found l_v decreased (Gamma GLM: $\beta = -0.00794$, $p = 0.183$), along with speed (Gamma GLM: $\beta = -0.0136$, $p < 0.001$), suggesting search strategy relies on maintaining position within a resource cluster. Resource unpredictability has found to have no significant effect on movement, but further investigation is warranted in this domain. Overall, our findings highlight the importance of incorporating ecological constraints into predictive movement models to better capture biologically plausible dynamics. This framework provides a generalisable approach for predicting how environmental change may influence animal movement, with potential applications in conservation planning and broader relevance to understanding movement processes across various ecological systems.

Keywords: Ballistic motion; Evolution; Foraging; Search behaviour; Movement ecology; Allometry

1 Introduction

Encounter rates fundamentally shape interactions between entities and plays a critical role across disciplines from ecology to physics and chemistry. Specifically in the movement ecology framework, understanding how animals adjust their movement to change encounter rates proves significant. Random search models have shown that directed (i.e., ballistic) trajectories and faster movement speeds result in higher encounter rates than tortuous (i.e., diffusive) trajectories and slower movement speeds (Mizumoto et al., 2017; Tejedor et al., 2012; Visser & Kiørboe, 2006). Individuals who exhibit diffusive movement will repeatedly search the same areas, while directed movement allows for larger areas of search within the same timespan. Because encounter rates determine resource acquisition, energetic demands predict that movement should scale predictable with body size (Brown et al., 2004). Indeed, both plankton and mammals alike vary in search strategy across body sizes, implicitly conferring differential encounter rates (Abe & Shimada, 2015; Noonan et al., 2023; Visser & Kiørboe, 2006). Thus, encounter theory provides a framework which ecologists can use to understand how movement ties individual fitness to population-level effects.

Despite increasing sophistication of these statistical models, their foundational assumptions limit their ecological realism. A persistent limitation is the decoupling of movement speed and persistence. Models investigating the effect of ballistic movement on encounter

rates commonly assume constant velocity (Locsei, 2007; Nolting et al., 2015; Tejedor et al., 2012), while models that do vary speed typically treat path geometry as a separately optimized parameter (Visser, 2007). A prime example is the canonical correlated random walk (CRW), which assumes step length and turning angle to be drawn from different statistical distributions (Kareiva & Shigesada, 1983). More generally, different movement models can all be reduced to the same characteristic scales (Gurarie & Ovaskainen, 2011), this approach collapses multiple trait combinations, making the mechanistic parameters difficult to identify. Furthermore, models routinely exclude evolutionary dynamics or assume movement is cost-free, and often omit processes such as range-residency and perceptual range that strongly influence encounter rates (Fagan et al., 2017; Martinez-Garcia et al., 2020; Nolting et al., 2015). These assumptions collectively obscure the mechanisms generating movement and limit the ability to predict ecologically realistic strategies.

However animals are not free to move without energetic constraints (Pontzer, 2025), and therefore movement strategies influence fitness outcomes (Viswanathan et al., 2011). In practice, animal movement is defined by i) movement speeds, and ii) directional persistence (Viswanathan et al., 2011). This suggests that these traits can be optimized either independently or in conjunction with one another, with respect the conditions of the landscape the individual is subject to. Since encounter rates with resources are determined by movement behaviour, optimal foraging theory (Garcia et al., 2007) predicts that selection should favour movement strategies that optimize encounters (Schooley & Wiens, 2003; Viswanathan et al., 2011; Wilson et al., 2015). Thus, the strength of bottom-up pressure may influence search strategy, such that dense resources should select for more diffusive movement, while sparse environments should lean towards more ballistic movement (López-López et al., 2013). However, both strategies have costs; overly ballistic search strategies may overshoot clustered resources (Bartumeus et al., 2014), while overly diffusive movement presents the risk that no resources will be found (Bartumeus et al., 2014; Dorfman et al., 2022). This presents a crucial trade-off, as animals navigate heterogeneous landscapes and must balance these

trade-offs with imperfect knowledge.

Considering the fitness trade-offs inherent to movement strategy, movement itself can be viewed as a composition of heritable traits under selective pressure. Over evolutionary timescales, these pressures may approach an equilibrium shaped by ecological constraints (Briffa & Lane, 2019; Putnins & Androulakis, 2021). Such a strategy, when common, can be resistant to invasion by rare alternative strategies, and is thus termed an evolutionarily stable strategy (ESS) (Briffa & Lane, 2019). For traditional movement models, the separate parametrization of movement speed and persistence limits inference about joint evolution (Gurarie et al., 2017). However, selective pressure acts on combinations of traits (Frank, 2025), highlighting that the decoupling of parameters can inhibit the ability to capture trade-offs and gives an incomplete ecological interpretation. While emerging work in cellular movement demonstrates these traits are intrinsically coupled (K. Chen & Qin, 2021; Shaejabi et al., 2020), this provides little insight into how these traits behave in terrestrial systems. For example, increasing speed may yield higher encounters, but imposes greater energetic costs, suggesting diminishing returns in fitness. Therefore, the relationship between movement strategy and encounter rates in terrestrial systems represents an underexplored area, warranting further investigation.

Importantly, bottom-up pressure strongly influences the fitness landscape for movement, shaping search strategies. As a result of human-induced rapid environmental change (HIREC), landscapes are becoming increasingly sparse, fragmented, and unpredictable (Sih, 2013). Environmental variability can generate fluctuating selection shifting towards instability of optima, implying that strategies stabilized under constant conditions may not persist under variable environments (Clark-Wolf et al., 2024). Landscape patchiness is an important influence in foraging strategy (Nauta et al., 2022). However, it is less understood how directional persistence and movement speed interact under stochasticity, and how this interaction influences search strategy. Accordingly, a framework that explicitly links movement parameters to environmental conditions is critical to understand animal search strategy and

guide conservation practice.

However, experiments on movement under varying resource availability at the landscape scale are not feasible, and movement data at evolutionary timescales are not directly observable. Thus, by utilising computational methodologies to investigate selection on search strategy under various landscape scenarios, we gain critical insight into how organisms adapt their movement. Here we present an individual-based model which simulates movement over generations and quantifies successful strategies across various environmental scenarios by offspring production. We use the Ornstein-Uhlenbeck with Foraging (OUF) model to simulate movement (Fleming et al., 2014), a mathematically tractable and empirically grounded variant of the Ornstein-Uhlenbeck (OU) stochastic process, which accounts for range-residency, as well as positional and velocity autocorrelation timescales (Noonan et al., 2019). We additionally integrate biological constraints on movement, through metabolic scaling and speed-dependent costs, to provide insights into the trade-offs faced by organisms between search strategy and energetic expenditure. We found that evolutionarily stable strategies emerge over time. Our results show that movement strategy changed based on environmental conditions, such that increased density facilitated faster, more diffusive movement, resource clustering facilitated slower, more diffusive movement, and caloric unpredictability facilitated faster, more ballistic movement.

2 Methods

The Model

A complete, detailed model description, following the ODD (Overview, Design Concepts, Details) protocol (Grimm, 2020; Grimm et al., 2006, 2010) is provided as supplementary. A visual description of the model is presented in Figure 1.

The basis of the presented model lies in encounter theory, where more persistent movement yields greater encounter rates. The purpose of this model is to determine how mammalian search strategies evolve and stabilise over evolutionary timescales. In order to quan-

tify search strategy, the ballistic length scale, l_v , describes the average distance over which directional movement persists, defined as a function of the spatial variance of an animal's movement (σ_p , in m^2), and their position and velocity autocorrelation timescales (τ_p and τ_v respectively, in sec), given by

$$l_v = \sqrt{\frac{\tau_v}{\tau_p}} \times \sigma \quad (1)$$

This measure gives a quantitative measure of an individual's search strategy.

The model consists of two main entities, the animals and the landscape. The animal entity represents an individual herbivorous mammal, which is characterised by its body mass, velocity and positional autocorrelation timescales (τ_v and τ_p respectively, in sec), along with its spatial scale (σ_p). Initial values of τ_v and τ_p are drawn from a uniform distribution, ranging from 0 to twice the expected value from the allometric function. The τ_p and τ_v traits are able to evolve over generation time, with random noise added via a normal distribution. On the other hand, σ_p is held constant, as it directly influences the home range size. By keeping σ_p constant, we investigate how animals change movement within a confined home range in order to optimize encounter rates. Starting body mass remains fixed across generations.

The landscape entity is designed to represent a continuous field in which an animal entity can navigate. Points are distributed via a Matérn cluster point process and the landscape extent is defined as a function of the positional variance, such that

$$EXT = \sqrt{(-2 \times \log_{10}(0.001) \times \pi) \times \sigma} \quad (2)$$

The landscape is characterised by the x, y coordinates of points, representing patches of resources, parameterised by the desired number of points and the clustering of these points. We re-defined the Matérn clustering process to characterise landscapes by the degree

of clustering (μ) and the number of desired points in the landscape. For homogeneous landscapes, μ was set to 1 to mimic a Poisson point process, while increased values would result in higher degrees of clustering. The intensity (λ) was defined as $\lambda = n_{points}/EXT$, while the number of parent points (κ) was defined as the ratio of λ and μ . The radius for offspring points (r) was set at a constant value of 1. Landscapes are further described by the caloric value of each point, assigned in units of *kcal*. This can either be constant across the landscape or vary according to a Gamma distribution. For all simulations, the mean caloric value of patches was held at 4000, which was slightly above the bare minimum calories required to obtain one offspring given the model parameters.

The model operates at generational timescales, with individual movement simulated at smaller discrete time steps within a single generation. Generation duration was defined as one one-thousandth of the allometrically determined lifespan (see Table S1, Magalhães et al., 2007). Within a generation, an individual undergoes five key processes in the following order, i) movement simulation, ii) resource acquisition, iii) energetic calculations, iv) reproduction, and vi) selection. Movement simulation involves the designation of initial locations, movement parameters τ_v, τ_p, σ_p , and simulation using the OUF movement model. Movement is simulated on a discrete timescale, using a sampling interval of $\frac{1}{10}\tau_v$, as determined by sensitivity analyses (See Supplementary Material B). Resource acquisition is determined by encounters between the movement trajectory and resource patches, with detection occurring within the perceptual range, defined as $pr = \sigma \times 0.05$. Once a patch is encountered it is considered consumed and cannot be encountered again within that individual's lifespan. Energetic costs are subtracted from total resource gains, and reproduction occurs if resource surpass a specific threshold based on allometry. Traits of individuals who successfully produce offspring are pooled and the next generation is sampled from this pool, such that the population size remains constant throughout the simulation. The resource landscape is reset at the beginning of each generation.

We defined energetic costs using empirically derived allometric relationships, focusing on

174 terrestrial mammalian species. Metabolic costs were calculated using the field metabolic
 175 rate for herbivorous mammals from Nagy, 1987, as shown in Equation 3. Energetic costs
 176 were calculated as a function of speed and body size, given by Equation 4 from Taylor et al.,
 177 1982. Both were converted to *kcal/s* and then multiplied by the generation duration time
 178 to get total energy expenditure.

$$\log_{10}(E_{\text{metabolic}}) = 0.774 + 0.727 \log_{10}(M) \quad (3)$$

$$E_{\text{transport-rate}} = \frac{(10.7M^{-0.316} \times v + 6.03M^{-0.303}) \times M}{1000} \quad (4)$$

179 Reproductive success was calculated using the framework described in the DEBKiss
 180 (Jager et al., 2013), such that 80 percent of energetic gains were allocated to somatic main-
 181 tenance and the remaining 20 percent were allocated to reproduction. Offspring costs were
 182 calculated using allometric scaling relationships from Blueweiss et al., 1978, with a conver-
 183 sion to dry assimilates with the assumption that neonates are composed of 70 percent wet
 184 material Young et al., 2021, as simplified in Equation 5. All allometric scaling relationships
 185 can be found in Table S1.

$$[\text{offspring}] = \frac{0.04 \times E_{\text{net}}^{0.08}}{0.02425 \times 0.16^{0.92}} \quad (5)$$

186 Environmental Scenarios

187 We present the findings for three key of environmental scenarios; resource density, patch
 188 spatial distribution, and patch caloric variance, as summarised in Table 1. Simulations were
 189 all conducted for a population of 400 animals, each with a body size of 105.5 kilograms.

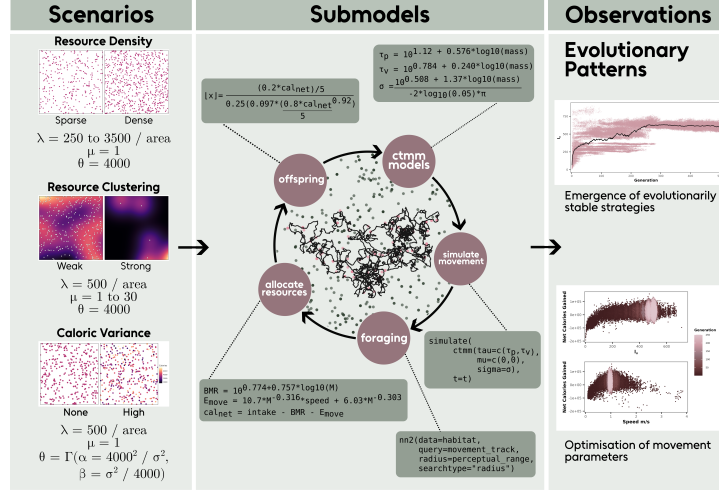


Figure 1: Visual overview of the model. During initialization animal entities are informed by allometric scaling relationships and a landscape scenario is created. Upon starting the simulation, animal entities generate movement parameters and their movement trajectory is simulated. Animals then forage, allocate resources, and ultimately produce offspring. This is repeated until the simulation is terminated at 500 generations (See supplementary material A for further details).

Each scenario was tested in isolation, such that resource density simulations were tested with homogeneous distribution of points, spatial distributions were tested with a constant number of points, and caloric variance was testing in a homogeneously distributed landscape with a constant number of points. A single simulation with 500 patches, no caloric variance, and no patch clustering was used as a baseline and appears in all 3 analyses.

Statistical Analyses

To quantify outcomes across environmental scenarios, each simulation was summarized by the mean l_v and speed over the final 10 generations, after population dynamics has stabilized. We then fit generalized linear models (GLMs) and non-linear least squares (NLS) to assess relationships between movement parameters and environmental variables. We used the NLS approach only for the relationship between resource density and speed in order to better characterise the saturating trend that a GLM could not capture. All other analyses used GLMs with a Gamma error distribution and a log link.

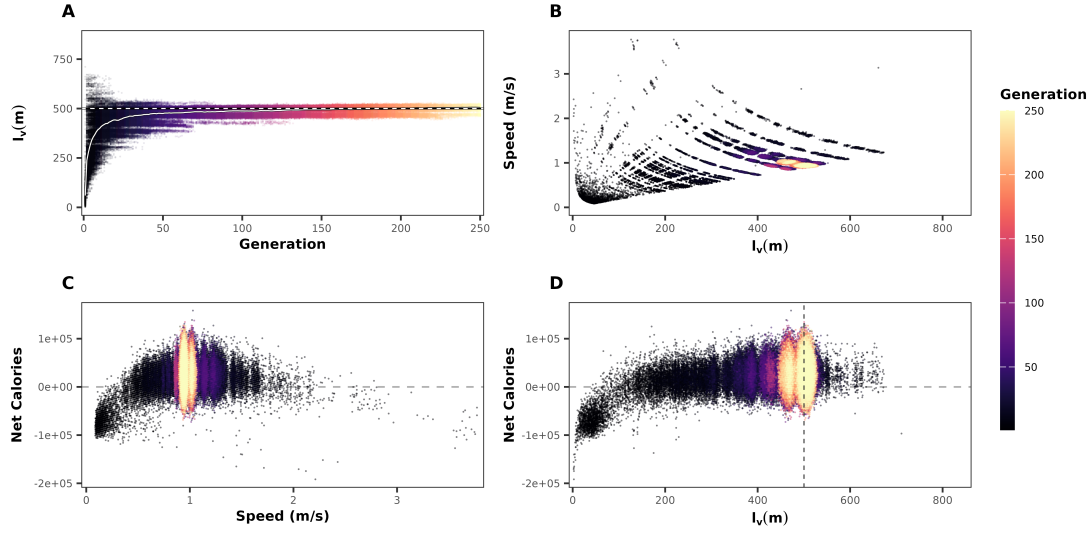


Figure 2: Evolution of movement strategy in a homogeneous landscape (250 patches, no clustering, no caloric variance). (A) Ballistic length scale (l_v) increases over generations (dashed line indicates stabilized value). (B) l_v and speed do not evolve independently. (C) Energy gained is optimized with respect to speed. (D) Energy gained is maximized at an intermediate, stabilized value of l_v . Points are coloured by generation. Evolutionarily stable strategies involve the optimization of both traits.

Table 1: Environmental Scenarios Tested

Scenario Component	Parameter Description	Values Tested	Number of Levels
Number of patches	Total number of food patches on the landscape	250 - 3500	82
Patch clustering	Degree of spatial clustering of patches (μ parameter controlling aggregation)	0 - 30	59
Patch calorie variance	Calorie Coefficient of Variation	0 - 5.25	31

3 Results

Our results highlight that evolutionarily stable strategies emerge for l_v over generation time (Figure 2A). We also found that l_v and movement showed a collinear relationship (Figure 2B). There was a narrow optimal range of movement speeds which maximized net caloric gain and ultimately fitness (Figure 2C). Finally, there was a large range of values in which l_v yielded maximal net caloric gain (Figure 2D).

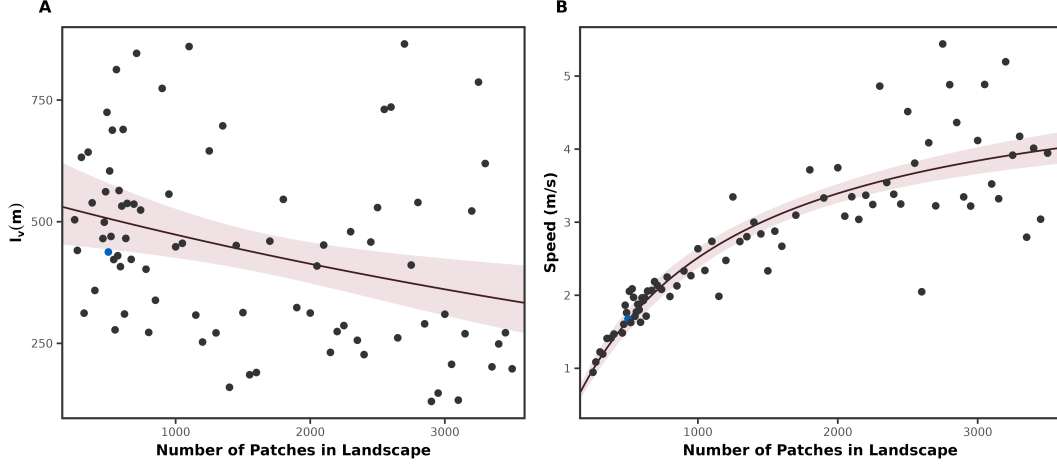


Figure 3: ESS in homogeneous landscapes with increasing density. (A) Stabilised ballistic length scale (l_v) decreased with resource density. (B) Stabilised speed (m/s) increases with resource density to a saturation point. The blue point indicates parametrization that is consistent across all scenarios.

Resource Density. We tested the population trends of movement strategy across a gradient of resource densities, ranging from 250 patches to 3500 patches available in the entire landscape. We found that, in general, l_v decreased with increasing resource density (Figure 3A, Gamma GLM, log link: $\beta = -1.35 \times 10^{-4}$, $se = 4.63 \times 10^{-5}$, $t = -2.93$, $p = 0.00448$). In contrast, speed increased with resource density to a saturation point (Figure 3B, Nonlinear least squares (Michaelis-Menten): $a = 5.22$, $se = 0.283$, $t = 18.480$, $p < 0.001$, $b = 1079$, $se = 147$, $t = 7.33$, $p < 0.001$). Variance around the l_v trend was notably higher than that observed for speed prior to saturation (3).

Resource Clustering. To evaluate the change in movement strategy as a result of resource spatial distribution, we ran simulations along a gradient of resource clustering. We started with a homogeneous Poisson point process and increased clustering following a Matérn clustering process. Our results show that l_v decreased insignificantly with increasing resource clustering (Figure 4A, Gamma GLM, log link: $\beta = -0.00794$, $se = 0.00589$, $t = -1.35$, $p = 0.183$). Speed decreased with increasing resource clustering (Figure 4B, Gamma GLM, log link: $\beta = -0.0136$, $se = 0.00186$, $t = -7.37$, $p < 0.001$). Similarly to the density result, variance around l_v was greater than speed.

Resource Unpredictability. Increasing the coefficient of variance (CoV) of caloric

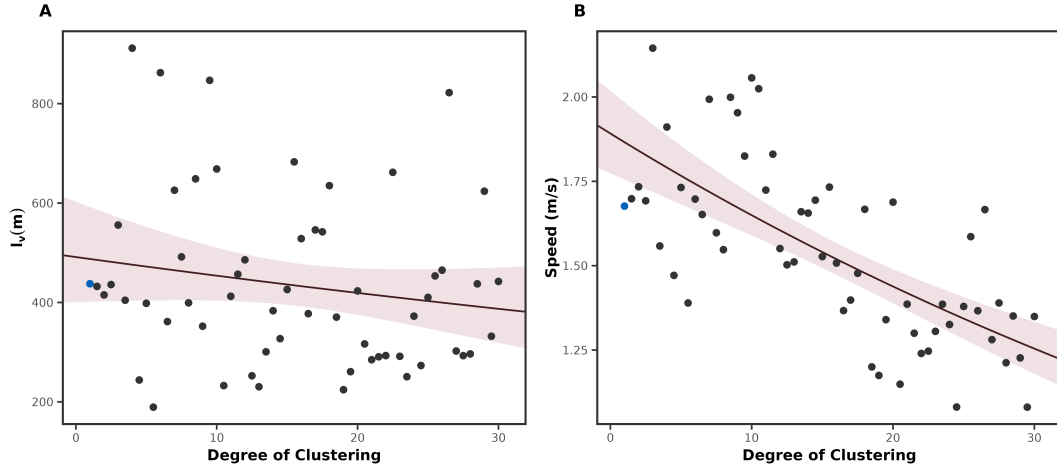


Figure 4: ESS in homogeneous landscapes with increased resource clustering. (A) Stabilised ballistic length scale (l_v) decreased with resource clustering. (B) Stabilised speed (m/s) decreased with resource clustering. The blue point indicates parametrization that is consistent across all scenarios.

value, we did not find statistically significant changes in movement. We saw that l_v increased with resource unpredictability (Figure 5A, Gamma GLM, log link: $\beta = 0.0415$, $se = 0.0288$, $t = 1.44$, $p = 0.161$) along with speed (Figure 5B, Gamma GLM, log link: $\beta = 0.00441$, $se = 0.00790$, $t = 0.558$, $p = 0.581$).

4 Discussion

Evolutionarily stable strategies in movement require optimization of l_v and speed, with optimal strategies varying with environmental conditions. Previous models have applied unrealistic assumptions which poorly represent terrestrial mammal movement patterns. Further, optimization-based approaches neglect evolutionary trade-offs that constrain real populations (Bull & Wang, 2010). By implementing the OUF stochastic process combined with allometric constraints on energetics, we present the first framework evaluating evolutionary dynamics of movement-mediated encounter rates in terrestrial systems. These results highlight how movement strategies shift under three specific environmental pressures: i) resource density, ii) spatial clustering, and, iii) patch quality variance. All of these scenarios are directly altered by anthropogenic climate change (Coristine & Kerr, 2011; Haddad et al., 2015; Kunegel-Lion et al., 2022), and represent key areas of interest for species management.

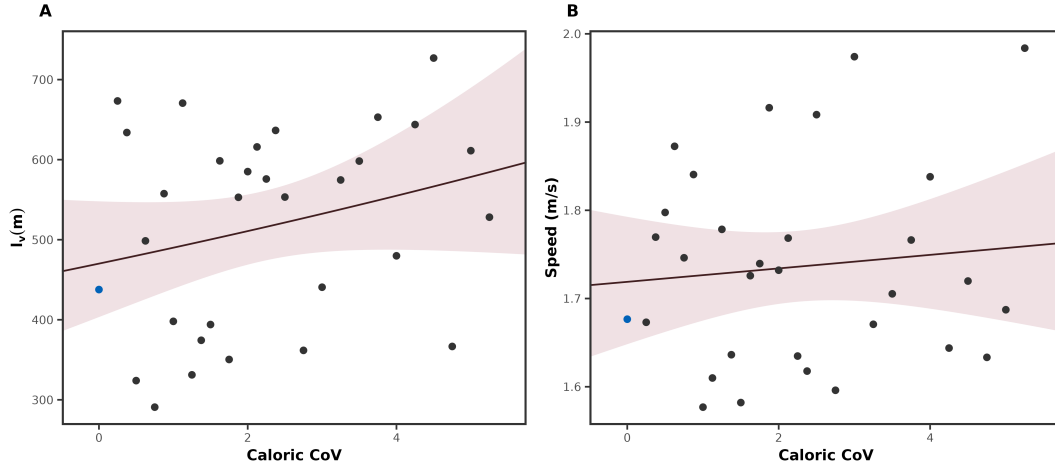


Figure 5: ESS in homogeneous landscapes with increased caloric variance. (A) Stabilised ballistic length scale (l_v) increased with caloric CoV. (B) Stabilised speed (m/s) increased with caloric CoV. The blue point indicates parametrization that is consistent across all scenarios.

For animals in all environmental scenarios, we found that there was a clear optimal range of movement speeds which maximize the net caloric gains, with this range shifting depending on the landscape scenario tested. These results reflect direct energetic trade-offs, as slow movement speeds reduce encounter rates below reproductive thresholds, while high speeds impose unsustainable energetic costs (Figure 2C). In contrast, l_v showed broad optimization, saturating at intermediate values rather than peaking (Figure 2D). In general, speed faces strongly selective pressure than l_v , which is more flexible to optimization within the broad range. The collinear relationship between l_v and speed emerges from their shared dependence on velocity autocorrelation timescales, constraining the range of achievable movement strategies. Our results demonstrate that maximal fitness requires the joint optimization of speed and l_v based on environmental conditions, rather than independent tuning of each trait.

Aligning with previous work (Visser & Kiørboe, 2006), we found that l_v decreased with increasing resource availability (Figure 3A). When resources are abundant, localized diffusive search outperforms long-distance ballistic movement. Speed increased with resource density, to a saturation point (Figure 3B). These results provide two key insights: i) faster movement increases encounters as predicted (Visser, 2007), and ii) increased resource availability offsets

the energetic costs associated with increased movement speed. The saturation of movement speeds suggests that increased movement speeds are only beneficial to a point, at which energetic costs are prohibitive regardless of intake. The higher variance in l_v relative to speed reflects different constraint mechanisms, as speed is tightly constrained by direct energetic expenditure, while l_v is constrained indirectly through its interaction with speed.

Both l_v and speed decreased with increased resource clustering (Figure 4), contradicting existing theoretical predictions. In sparse, patchy environments, increased clustering shifts the search problem toward longer inter-patch movements, for which more ballistic motion is predicted to improve encounter rates (Grünbaum, 2012). Our results could be aligned with the regularity hypothesis, which states animals may remain in a food patch not because they are optimizing energetic gain, but due to preference of regular, predictable encounters with resources (Nepomniashchikh, 2003, as cited in Viswanathan et al., 2011). This would indicate that diffusive and slow strategies are more predictable for acquiring resources compared to ballistic strategies in highly fragmented landscapes. Another plausible explanation for the deviation from optimal foraging theory is the structural framework of the model. Reproduction was calculated cumulatively at the end each generation, rather than as a per time step mechanism. This allowed individuals who moved slow and diffusively to remain within a dense resource cluster and fully deplete it. These individuals would accumulate sufficient calories for reproduction, without the need to relocate, therefore favouring localized search rather than long-range exploration. In reality, animals must maintain energetic balance continuously and relocate following the depletion of resources (Auer et al., 2020), which should favour higher l_v in clustered landscapes to facilitate the movement between dispersed patches. In order to correct this, per time step energetic framework and mortality would force individuals to search for new clusters following depletion, theoretically favouring increased l_v and speed in patchy environments (Visser & Kiørboe, 2006).

Patch quality variance showed non-significant trends towards increased l_v and speed (Figure 5, $p > 0.05$ for both). This result likely reflects insufficient variance in our experimental

setup, as the tested coefficients of variation produced patch quality ranges of only ± 400 kcal around the mean. Theory predicts that high environmental unpredictability favours broader search (Viswanathan et al., 1999), but this requires patch quality distributions spanning the full viable range, including zero-value patches. Future iterations should specify that patch quality ranges from zero to well above the mean, and be more evenly distributed across that range. This change would impose stronger selective pressure on search strategies and likely reveal significant effects.

These emergent relationships are important to understanding the response of animal movement in the wake of climate change. It is well-established that population ranges shift in response to changing environmental conditions (I.-C. Chen et al., 2011; Mezzini et al., 2025), but how animals navigate within their home range is less understood. Human-induced rapid environmental change (HIREC) manifests through habitat and climate change (Sih, 2013), including decreased abundance (Cristine & Kerr, 2011), increased fragmentation (Haddad et al., 2015), and heightened unpredictability (Kunegel-Lion et al., 2022). Our results provide three specific predictions for how HIREC will alter movement strategies. First, habitat loss that reduces overall resource density should select for faster, more ballistic strategies, as animals must cover larger areas while managing energetic costs. These results complement the relationship between resource abundance and range size (Mezzini et al., 2025), as animals will require larger areas of space to conduct long-distance searches in sparse environments. Second, increased fragmentation may favour slower, more diffusive movement, potentially making animals more vulnerable to local resource depletion and less able to respond to disturbances. If animals become adapted to highly restrictive search, they may become more sensitive to stochastic events (e.g., extreme weather events) and fail to relocate efficiently when required. Finally, increased environmental stochasticity should drive faster, more ballistic search, though our current results suggest that this prediction requires further investigation.

For this model, we implemented assumptions in order to simplify the framework and

313 increase computational feasibility. **Temporal discretization.** Although the OUF process
 314 operates in continuous time, simulation required discretization at a specified interval. In
 315 order to ensure minimal impact of this discretization, we utilised the velocity autocorrela-
 316 tion timescale and sensitivity analyses to ensure that all potential patches along a movement
 317 track were consumed. Based on these analyses, we found the a sampling interval of 38 sec-
 318 onds, which corresponds to $\Delta t = \tau_v/10$, was sufficient to minimize discretization artifacts
 319 (see Supplementary Material B). **Perceptual range.** We defined the perceptual range as a
 320 fixed radius of $pr = \sigma \times 0.05$, corresponding to 57 m for our 105.5 kilogram individual. We
 321 assumed patch occurrence within this range lead to consumption and did not include the
 322 role which senses have to locating resources. Thus, our model omits visual, olfactory, and
 323 auditory cues that extend effective perception beyond the physical range in many species
 324 (Fagan et al., 2017). This causes our model to underestimate true encounter rates in species
 325 with enhanced sensory abilities, and does not account for reactive changes in movement
 326 strategy. Additionally, we did not consider handling time within the framework, thus over-
 327 estimating the rate and efficiency of consumption. **Cognitive and social factors.** The
 328 model omits spatial memory (return to known productive areas (Ranc et al., 2021)), learning
 329 (improved search efficiency over time(Ranc et al., 2021)), conspecific competition (resource
 330 depletion by others (Smolla et al., 2015)), and predation risk (search-safety trade-offs (Kie,
 331 1999; Noonan et al., 2023)). Each of these factors is known to shape movement strategies in
 332 natural systems and represents an area where the model could be extended. **Reproduction**
 333 **framework.** We defined reproduction as a cumulative process occurring at the end of the
 334 simulation, rather than a per time step balance. This results in model bias towards cluster ex-
 335 ploitation versus long-term sustainable search, producing the counter-intuitive results in the
 336 clustering analysis (see above). Implementing continuous reproduction and energy budgets
 337 should correct this. **Multivariate environmental change.** We tested resource density,
 338 clustering, and variance independently, but environmental change occurs upon maybe axes
 339 simultaneously (Opdam & Wascher, 2004). Interactive effects such as habitat loss reduc-

ing density and increasing fragmentation remain unexplored and may produce non-additive fitness consequences.

5 Conclusions

This work presents the first evolutionary model linking terrestrial mammalian movement processes to population-level encounter rates and fitness. By using empirically validation OUF movement with allometric energetic constraints, we demonstrate that joint optimization of speed and directional persistence is necessary for maximizing encounter rates and that optimal strategies shift predictably with resource density and distribution. These findings have direct implications for conservation practice. Protected area design should account for how habitat degradation alters movement strategies, not just species' metabolic requirements. Connectivity planning should consider resource spatial distribution, rather than just total abundance, as clustered restoration may be less effective than distributed efforts for specific species. Despite current limitations, this framework provides a mechanistic foundation for predicting movement responses to environmental change. Future work should investigate intra- and inter-specific interactions, along with multivariate environmental set-ups. Additionally, the use of system specific framework could provide quantitative assessments of movement strategy, helping to design effective conservation interventions in rapidly changing landscapes.

Supplementary Materials

Table S1: Allometric scaling equations for movement parameters. Note that body mass, M , is needed in grams (g).

Parameter	Equation	Source
Field Metabolic Rate	$10^{0.774} \times M^{0.727}$	Nagy, 1987
Movement costs	$10.7 \times M^{-0.316} \times velocity + 6.03 \times (\frac{M}{1000})^{-0.303}$	Taylor et al., 1982
Lifespan	$4.88 \times M^{0.153}$	Magalhães et al., 2007

Parameter	Equation	Source
95% Home Range	$HR_{95} = 10^{0.5079} \times M^{1.3721}$	Noonan et al., 2020
Variance in position	$\sigma_p^2 = \frac{HR_{95}}{-2 \ln(0.05)\pi}$	Noonan et al., 2020
Position autocorrelation	$\tau_p = 10^{1.2994} \times M^{0.8129}$	Noonan et al., 2020
Position autocorrelation	$\tau_v = 10^{-1.3652} \times M^{0.7871}$	Noonan et al., 2020
Offspring requirements	$0.02425 \times M^{0.92}$	Blueweiss et al., 1978; Young et al., 2021

Data Availability

All scripts used for this project are available on [GitHub](#).

References

- Abe, M. S., & Shimada, M. (2015). Lévy Walks Suboptimal under Predation Risk. *PLOS Computational Biology*, 11(11), e1004601. <https://doi.org/10.1371/journal.pcbi.1004601>
- Auer, S. K., Solowey, J. R., Rajesh, S., & Rezende, E. L. (2020). Energetic mechanisms for coping with changes in resource availability. *Biology Letters*, 16(11), 20200580. <https://doi.org/10.1098/rsbl.2020.0580>
- Bartumeus, F., Raposo, E. P., Viswanathan, G. M., & da Luz, M. G. E. (2014). Stochastic Optimal Foraging: Tuning Intensive and Extensive Dynamics in Random Searches. *PLoS ONE*, 9(9), e106373. <https://doi.org/10.1371/journal.pone.0106373>
- Blueweiss, L., Fox, H., Kudzma, V., Nakashima, D., Peters, R., & Sams, S. (1978). Relationships between body size and some life history parameters. *Oecologia*, 37(2), 257–272. <https://doi.org/10.1007/BF00344996>
- Briffa, M., & Lane, S. M. (2019, January). Signals in Conflict Resolution: Conventional Signals, Aggression and Territoriality. In J. C. Choe (Ed.), *Encyclopedia of Animal Behavior (Second Edition)* (pp. 531–538). Academic Press. <https://doi.org/10.1016/B978-0-12-809633-8.90701-5>
- Brown, J. H., Gillooly, J. F., Allen, A. P., Savage, V. M., & West, G. B. (2004). Toward a Metabolic Theory of Ecology. *Ecology*, 85(7), 1771–1789. <https://doi.org/10.1890/03-9000>
- Bull, J. J., & Wang, I.-N. (2010). Optimality models in the age of experimental evolution and genomics. *Journal of evolutionary biology*, 23(9), 1820–1838. <https://doi.org/10.1111/j.1420-9101.2010.02054.x>
- Chen, I.-C., Hill, J. K., Ohlemüller, R., Roy, D. B., & Thomas, C. D. (2011). Rapid Range Shifts of Species Associated with High Levels of Climate Warming. *Science*, 333(6045), 1024–1026. Retrieved April 21, 2026, from <https://www.jstor.org/stable/27978495>

- Chen, K., & Qin, K.-R. (2021). Random Walks of a Cell With Correlated Speed and Persistence Influenced by the Extracellular Topography. *Frontiers in Physics*, 9. <https://doi.org/10.3389/fphy.2021.719293>
- Clark-Wolf, T. J., Boersma, P. D., Plard, F., Rebstock, G. A., & Abrahms, B. (2024). Increasing environmental variability inhibits evolutionary rescue in a long-lived vertebrate. *Proceedings of the National Academy of Sciences*, 121(34), e2406314121. <https://doi.org/10.1073/pnas.2406314121>
- Coristine, L. E., & Kerr, J. T. (2011). Habitat loss, climate change, and emerging conservation challenges in Canada. *Canadian Journal of Zoology*, 89(5), 435–451. <https://doi.org/10.1139/z11-023>
- Dorfman, A., Hills, T. T., & Scharf, I. (2022). A guide to area-restricted search: A foundational foraging behaviour. *Biological Reviews of the Cambridge Philosophical Society*, 97(6), 2076–2089. <https://doi.org/10.1111/brv.12883>
- Fagan, W. F., Gurarie, E., Bewick, S., Howard, A., Cantrell, R. S., & Cosner, C. (2017). Perceptual Ranges, Information Gathering, and Foraging Success in Dynamic Landscapes. *The American Naturalist*, 189(5), 474–489. <https://doi.org/10.1086/691099>
- Fleming, C., Calabrese, J., Mueller, T., Olson, K., Leimgruber, P., & Fagan, W. (2014). From Fine-Scale Foraging to Home Ranges: A Semivariance Approach to Identifying Movement Modes across Spatiotemporal Scales. *The American naturalist*, 183, E154–67. <https://doi.org/10.1086/675504>
- Frank, S. A. (2025). Natural selection at multiple scales. *Evolution*, 79(7), 1166–1184. <https://doi.org/10.1093/evolut/qpaf037>
- Garcia, R., Moss, F., Nihongi, A., Strickler, J. R., Göller, S., Erdmann, U., Schimansky-Geier, L., & Sokolov, I. M. (2007). Optimal foraging by zooplankton within patches: The case of *Daphnia*. *Mathematical Biosciences*, 207(2), 165–188. <https://doi.org/10.1016/j.mbs.2006.11.014>

- Grimm, V. (2020). The ODD protocol: An update with guidance to support wider and more consistent use. *Ecological Modelling*, 428, 109105. <https://doi.org/10.1016/j.ecolmodel.2020.109105>
- Grimm, V., Berger, U., Bastiansen, F., Eliassen, S., Ginot, V., Giske, J., Goss-Custard, J., Grand, T., Heinz, S. K., Huse, G., Huth, A., Jepsen, J. U., Jørgensen, C., Mooij, W. M., Müller, B., Pe'er, G., Piou, C., Railsback, S. F., Robbins, A. M., ... DeAngelis, D. L. (2006). A standard protocol for describing individual-based and agent-based models. *Ecological Modelling*, 198(1), 115–126. <https://doi.org/10.1016/j.ecolmodel.2006.04.023>
- Grimm, V., Berger, U., DeAngelis, D. L., Polhill, J. G., Giske, J., & Railsback, S. F. (2010). The ODD protocol: A review and first update. *Ecological Modelling*, 221(23), 2760–2768. <https://doi.org/10.1016/j.ecolmodel.2010.08.019>
- Grünbaum, D. (2012). The logic of ecological patchiness. *Interface Focus*, 2(2), 150–155. <https://doi.org/10.1098/rsfs.2011.0084>
- Gurarie, E., & Ovaskainen, O. (2011). Characteristic Spatial and Temporal Scales Unify Models of Animal Movement. *AMERICAN NATURALIST*, 178(1), 113–123. <https://doi.org/10.1086/660285>
- Gurarie, E., Fleming, C. H., Fagan, W. F., Laidre, K. L., Hernández-Pliego, J., & Ovaskainen, O. (2017). Correlated velocity models as a fundamental unit of animal movement: Synthesis and applications. *Movement Ecology*, 5(1), 13. <https://doi.org/10.1186/s40462-017-0103-3>
- Haddad, N. M., Brudvig, L. A., Clobert, J., Davies, K. F., Gonzalez, A., Holt, R. D., Lovejoy, T. E., Sexton, J. O., Austin, M. P., Collins, C. D., Cook, W. M., Damschen, E. I., Ewers, R. M., Foster, B. L., Jenkins, C. N., King, A. J., Laurance, W. F., Levey, D. J., Margules, C. R., ... Townshend, J. R. (2015). Habitat fragmentation and its lasting impact on Earth's ecosystems. *Science Advances*, 1(2), e1500052. <https://doi.org/10.1126/sciadv.1500052>

- Jager, T., Martin, B. T., & Zimmer, E. I. (2013). DEBkiss or the quest for the simplest generic model of animal life history. *Journal of Theoretical Biology*, 328, 9–18. <https://doi.org/10.1016/j.jtbi.2013.03.011>
- Kareiva, P. M., & Shigesada, N. (1983). Analyzing insect movement as a correlated random walk. *Oecologia*, 56(2), 234–238. <https://doi.org/10.1007/BF00379695>
- Kie, J. G. (1999). Optimal Foraging and Risk of Predation: Effects on Behavior and Social Structure in Ungulates. *Journal of Mammalogy*, 80(4), 1114–1129. <https://doi.org/10.2307/1383163>
- Kunegel-Lion, M., Neilson, E. W., Mansuy, N., & Goodsman, D. W. (2022). Habitat quality does not predict animal population abundance on frequently disturbed landscapes. *Ecological Modelling*, 469, 109943. <https://doi.org/10.1016/j.ecolmodel.2022.109943>
- Locsei, J. T. (2007). Persistence of direction increases the drift velocity of run and tumble chemotaxis. *Journal of Mathematical Biology*, 55(1), 41–60. <https://doi.org/10.1007/s00285-007-0080-z>
- López-López, P., Benavent-Corai, J., García-Ripollés, C., & Urios, V. (2013). Scavengers on the Move: Behavioural Changes in Foraging Search Patterns during the Annual Cycle. *PLOS ONE*, 8(1), e54352. <https://doi.org/10.1371/journal.pone.0054352>
- Magalhães, J. P. d., Costa, J., & Church, G. M. (2007). An Analysis of the Relationship Between Metabolism, Developmental Schedules, and Longevity Using Phylogenetic Independent Contrasts. *The Journals of Gerontology: Series A*, 62(2), 149–160. <https://doi.org/10.1093/gerona/62.2.149>
- Martinez-Garcia, R., Fleming, C. H., Seppelt, R., Fagan, W. F., & Calabrese, J. M. (2020). How range residency and long-range perception change encounter rates. *Journal of Theoretical Biology*, 498, 110267. <https://doi.org/10.1016/j.jtbi.2020.110267>
- Mezzini, S., Fleming, C. H., Medici, E. P., & Noonan, M. J. (2025). How resource abundance and resource stochasticity affect organisms’ range sizes. *Movement Ecology*, 13(1), 20. <https://doi.org/10.1186/s40462-025-00546-5>

- Mizumoto, N., Abe, M. S., & Dobata, S. (2017). Optimizing mating encounters by sexually dimorphic movements. *Journal of The Royal Society Interface*, 14(130), 20170086. <https://doi.org/10.1098/rsif.2017.0086>
- Nagy, K. A. (1987). Field Metabolic Rate and Food Requirement Scaling in Mammals and Birds. *Ecological Monographs*, 57(2), 111–128. <https://doi.org/10.2307/1942620>
- Nauta, J., Simoens, P., Khaluf, Y., & Martinez-Garcia, R. (2022). Foraging behaviour and patch size distribution jointly determine population dynamics in fragmented landscapes. *Journal of The Royal Society Interface*, 19(191), 20220103. <https://doi.org/10.1098/rsif.2022.0103>
- Nolting, B. C., Hinkelman, T. M., Brassil, C. E., & Tenhumberg, B. (2015). Composite random search strategies based on non-directional sensory cues. *Ecological Complexity*, 22, 126–138. <https://doi.org/10.1016/j.ecocom.2015.03.002>
- Noonan, M. J., Fleming, C. H., Tucker, M. A., Kays, R., Harrison, A.-L., Crofoot, M. C., Abrahms, B., Alberts, S. C., Ali, A. H., Altmann, J., Antunes, P. C., Attias, N., Belant, J. L., Beyer Jr., D. E., Bidner, L. R., Blaum, N., Boone, R. B., Caillaud, D., de Paula, R. C., ... Calabrese, J. M. (2020). Effects of body size on estimation of mammalian area requirements. *Conservation Biology*, 34(4), 1017–1028. <https://doi.org/10.1111/cobi.13495>
- Noonan, M. J., Martinez-Garcia, R., Fleming, C. H., De Figueiredo, B. G., Ali, A. H., Attias, N., Belant, J. L., Beyer, D. E., Berteaux, D., Bidner, L. R., Boone, R., Boutin, S., Brito, J., Brown, M., Carter, A., Castellanos, A., Castellanos, F. X., Chitwood, C., Darlington, S., ... Calabrese, J. M. (2023). The search behavior of terrestrial mammals. *bioRxiv*, 2022.12.31.521874. <https://doi.org/10.1101/2022.12.31.521874>
- Noonan, M. J., Tucker, M. A., Fleming, C. H., Akre, T. S., Alberts, S. C., Ali, A. H., Altmann, J., Antunes, P. C., Belant, J. L., Beyer, D., Blaum, N., Böhning-Gaese, K., Cullen Jr., L., de Paula, R. C., Dekker, J., Drescher-Lehman, J., Farwig, N., Fichtel, C., Fischer, C., ... Calabrese, J. M. (2019). A comprehensive analysis of

autocorrelation and bias in home range estimation. *Ecological Monographs*, 89(2),
e01344. <https://doi.org/10.1002/ecm.1344>

Opdam, P., & Wascher, D. (2004). Climate change meets habitat fragmentation: Linking
landscape and biogeographical scale levels in research and conservation. *Biological
Conservation*, 117(3), 285–297. <https://doi.org/10.1016/j.biocon.2003.12.008>

Pontzer, H. (2025). The energetics of movement, from exercise to ecology and evolution.
Journal of Experimental Biology, 228(Suppl_1), JEB247988. <https://doi.org/10.1242/jeb.247988>

Putnins, M., & Androulakis, I. P. (2021). Self-selection of evolutionary strategies: Adaptive
versus non-adaptive forces. *Heliyon*, 7(5), e06997. <https://doi.org/10.1016/j.heliyon.2021.e06997>

Ranc, N., Moorcroft, P. R., Ossi, F., & Cagnacci, F. (2021). Experimental evidence of
memory-based foraging decisions in a large wild mammal. *Proceedings of the Na-
tional Academy of Sciences of the United States of America*, 118(15), e2014856118.
<https://doi.org/10.1073/pnas.2014856118>

Schooley, R. L., & Wiens, J. A. (2003). Finding habitat patches and directional connec-
tivity [eprint: <https://nsojournals.onlinelibrary.wiley.com/doi/pdf/10.1034/j.1600-0706.2003.12490.x>]. *Oikos*, 102(3), 559–570. <https://doi.org/10.1034/j.1600-0706.2003.12490.x>

Shaebani, M. R., Jose, R., Santen, L., Stankevicius, L., & Lautenschläger, F. (2020). Persistence-
Speed Coupling Enhances the Search Efficiency of Migrating Immune Cells. *Physical
Review Letters*, 125(26), 268102. <https://doi.org/10.1103/PhysRevLett.125.268102>

Sih, A. (2013). Understanding variation in behavioural responses to human-induced rapid
environmental change: A conceptual overview. *Special Issue: Behavioural Plasticity
and Evolution*, 85(5), 1077–1088. <https://doi.org/10.1016/j.anbehav.2013.02.017>

Smolla, M., Gilman, R. T., Galla, T., & Shultz, S. (2015). Competition for resources can
explain patterns of social and individual learning in nature. *Proceedings of the Royal*

Society B: Biological Sciences, 282(1815), 20151405. <https://doi.org/10.1098/rspb.2015.1405>

Taylor, C. R., Heglund, N. C., & Maloiy, G. M. O. (1982). Energetics and mechanics of terrestrial locomotion. I. Metabolic energy consumption as a function of speed and body size in birds and mammals. *Journal of Experimental Biology*, 97(1), 1–21. <https://doi.org/10.1242/jeb.97.1.1>

Tejedor, V., Voituriez, R., & Bénichou, O. (2012). Optimizing Persistent Random Searches. *Physical Review Letters*, 108(8), 088103. <https://doi.org/10.1103/PhysRevLett.108.088103>

Visser, A. W. (2007). Motility of zooplankton: Fitness, foraging and predation. *Journal of Plankton Research*, 29(5), 447–461. <https://doi.org/10.1093/plankt/fbm029>

Visser, A. W., & Kiørboe, T. (2006). Plankton motility patterns and encounter rates. *Oecologia*, 148(3), 538–546. <https://doi.org/10.1007/s00442-006-0385-4>

Viswanathan, G. M., Buldyrev, S. V., Havlin, S., da Luz, M. G. E., Raposo, E. P., & Stanley, H. E. (1999). Optimizing the success of random searches. *Nature*, 401(6756), 911–914. <https://doi.org/10.1038/44831>

Viswanathan, G. M., da Luz, M. G. E., Raposo, E. P., & Stanley, H. E. (2011). *The Physics of Foraging: An Introduction to Random Searches and Biological Encounters*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511902680>

Wilson, R. S., Husak, J. F., Halsey, L. G., & Clemente, C. J. (2015). Predicting the Movement Speeds of Animals in Natural Environments. *Integrative and Comparative Biology*, 55(6), 1125–1141. <https://doi.org/10.1093/icb/icv106>

Young, A., Brown, L. K., Ennis, S., Beattie, R. M., & Johnson, M. J. (2021). Total body water in full-term and preterm newborns: Systematic review and meta-analysis. *Archives of Disease in Childhood - Fetal and Neonatal Edition*, 106(5), 542. <https://doi.org/10.1136/archdischild-2020-321112>